

A pre-feasibility study for basalt ore extraction



2017

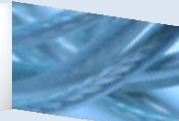


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1. Executive summary

The idea of the project is to extract the basalt ores, and carries out process required to convert it into many important products to be suitable for the local community needs and opening of several prospects for exporting. The basalt ores are located in various areas of the Kingdom and in large quantities in Mafraq governate especially in Bahrat ASham Northeast the Kingdom, and this exploitable amount is appropriate to establish a factory and quarry for production to supply the local market of basalt ores which is important because of its hardness and resistance to corrosion.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,039,204
Internal Rate of Return (IRR)	22.98%
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	850,004
Job Opportunities provided by the Project	11
Total Revenues (For 10 Years)	10,260,172
Total Operating Expenses (For 10 years)	6,150,937
Total Net Profit (For 10 years)	967,597

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Million distributed among 12 Governorates.

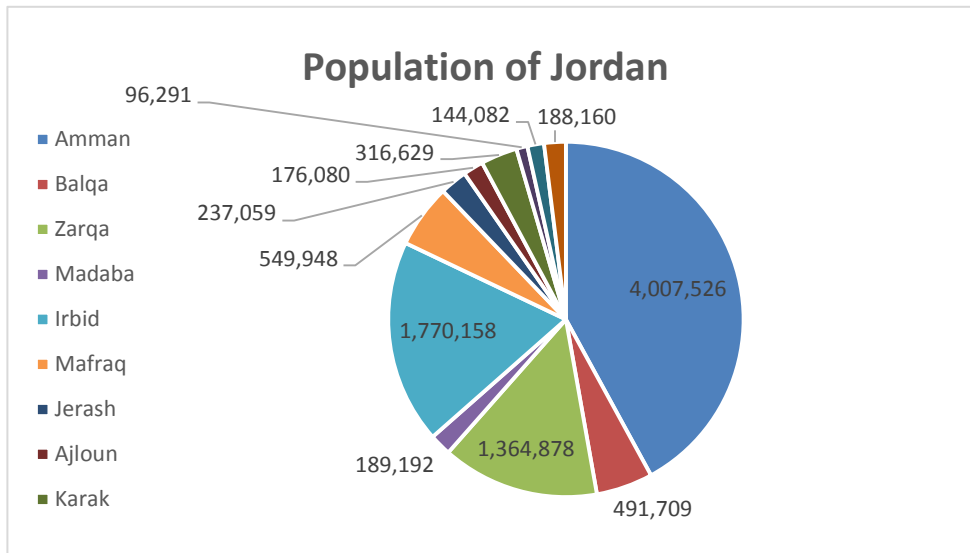
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

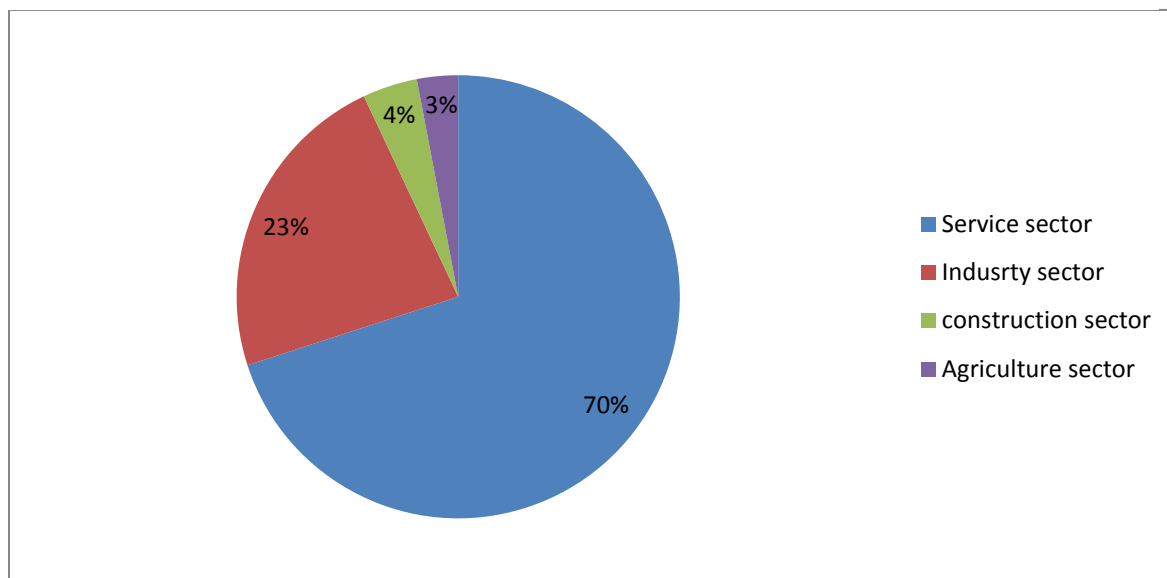
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the

Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally. The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and nonresidents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all-time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and

legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.

- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Mafraq:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) Of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 payable income tax and tax exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Mafraq governorate is classified as category C where investors get 60% tax exemption on income tax for 20 years from the date of investment.

3. Market analysis

Project description

The proposed project is to extract the basalt ores from Bahrat Asham areas which contains large reserves, the extracting and processing will be carried out for various industrial for many uses by crushing and sieving an ores to produce different particle sizes to be introduced to various industries such as rock wool industry (thermal insulators), it is proposed to establish the factory on 20 donum as a first step of the project, the area will be increased in the future based on market requirements.

Basalt

It is a volcanic igneous rocks containing less than 53% of silica (SiO₂). Due to the low content of silica in basalt rocks its decreases the viscosity (resistance to flow); therefore, basaltic lava can flow quickly and move easily. The low viscosity allows the volcanic gases to exit without generating enormous explosions. Among the common crystals in basalt: Olivine, Pyroxene and Plagioclase where the basalt explodes at temperature between 1100°C to 1250 °C, and the basaltic rocks are igneous rock and it texture is (glass) or

microscopic, because the crystals are solidifying near the Earth's surface and leads to aggregate the ions around the crystalline center to give very fine fabric. Where the basalt is heavy rocks and a very hard when it touched the surface are rough and very difficult to break it.

Basalt occurrences in Jordan

Northeast of the Kingdom; Wadi Sarhan areas, ALazraq, Safawi, east of Mafraq and Bahrat Sham. These rocks cover an area of 11,000 km² and extend to the length of 180 km and width of 50_70 km. One of these vast areas of the basaltic presence it is recommended to exploit the basalt which is located in the east of Mafraq because it is close to high way road.

Center of the kingdom: Ma'een areas, Mkawer, Zara.

South of the Kingdom: Tal Parma, Unaiyza mountain, Alaghira, Kobraha mountain and Al Attaath mountain.

Chemical composition

Basaltic rocks consists of a group of elements and oxides such as: SiO₂, Al₂O₃, Fe₂O₃, FeO, MgO, CaO, Na₂O and K₂O, in addition to other trace oxides, from these oxides and chemical compositions consists of rocks, the Ogyte is the essential element in the formation of basalt, followed by blgioclase, Alnivilin, olivine, bronzite and ortoclase .the ratios are various in the basaltic composition.

Table3 the major oxides of the basalt

Oxide	Percentage%
Fe ₂ O ₃	13.3 – 14.5
MnO	0.18 – 0.23
TiO ₂	2.8 – 3.2
CaO	9.8 – 11.4
K ₂ O	0.53 – 1.3
SiO ₂	4.2-4.0
Na ₂ O	0.61 – 2.5
Al ₂ O ₃	11.8 – 12.7
P ₂ O ₅	0.57 – 0.65
MgO	9.16 – 9.9

Source: Natural resources authority. (2006)

There are many differences in basalt composition which is due to the different structures, the table 5 below shows the chemical composition of the basalt

Table 4 the chemical composition of the basalt

Area	SiO ₂ %	Al ₂ O ₃ %	Fe ₂ O ₃ %	CaO%	MgO%	Na ₂ O%	K ₂ O%
Tal Parma, Unaiyza	40-43	11.8-	13.2-	9.9-	9.2-9.8	0.62-2.5	0.53-
mountain		12.7	14.3	11.8			1.3
Qaa' khanna	42.60	16.58	12.87	12.60	9.45	0.46	0.88

Natural resources authority (2006)

Physical properties:

The physical properties of the basaltic rocks are vary because the difference over the structures and textures and the temperature range of 1000 °C - 1200 ° C, and the viscosity of lava range between 103-610 Poise and increasing with decreasing the temperature and increasing the crystalline content. The surface tension range for the molten between 270-350 dyne / cm and the density of the molten ranging between 2.6 to 2.7 g / cm³ at 1200 ° C, while the crystalline rock in standard conditions (STP) and lack of voids ranging between 2.8 to 3.1 g / cm³. The value of the melting and crushing resistance ranging in cold rock between 1700 - 2200 atmospheric pressure, and the strength of cohesion in the rock itself ranges between 320-440 atmospheric pressure.

The electrical conductivity equal 100 for each cm approximately for a molten at 1200 ° C, while the thermal conductivity about $1-3 \times 10^{-3}$ price / cm °C of molten at 1200 ° C, heat capacity under constant pressure between 2 and 3.

The mineral composition and basaltic texture are varying with the difference of the cooling and chemical composition of the magma. In the slow cooling the crystals are grow to large sizes and take the form of an equilateral.

Crystals grows rapidly under the influence of cooling conditions which have a chemical composition differ than the slow cooling. As a result the basalt flow can contain different metals in different places with it cooling rates. The size of the crystals in basaltic rocks present the rate crystals, the large crystals maybe grown during a few hours or days during magma flow through the crust, otherwise many of the crystals maybe grown in weeks or years before the impulsion of it in a large reservoir of magma, which cools slowly because of the large area and slowly cooling due to the high ratio between the surface area and the volume.

Project objectives

- Exploitation millions of tons of basalt ore in eastern Mafraq, known as Bahrat Sham. The ores in this area covers about 11,000 km² and extends with length of 180 km and width of 50-70 km, which enables these minerals suitable for using in several industries.
- Exploiting and treatment will be resulted in getting basalt ores with high quality, the products of high purity will encourage the local industries to replace the foreign products by local products.
- Enhancing the country development by exploiting the local mineral resources by using (basaltic products) in different industries such as: Rockwool, molds, building stone and decoration.

Proposed product

The uses:

- Coating the outdoor walls and the floors of the building and decoration.
- Use the different sizes of basaltic rocks in concrete, bridges, dams and building raw materials.
- Used as aggregate in concrete.
- Used as a court in sidewalks and streets.
- Used as filler.
- In producing of industrial equipments with acidic and alkaline resistance.
- Used in basaltic plastic pipes.
- Used in insulator materials.
- Treat the remnants of industries.
- Prevent the chemical and industrial building from damaging, caused by oxidization.
- Produce the basaltic fabric.
- Produce the hard rods that used in construction.
- Produce basaltic Rockwool.
- Used in space and military industries.
- Used in composite materials such as: pollster, plastic used in cars body.

Targeted segments

The project will target different consumers such as:

- Al Motaheda Company for concretes
- Concorde company for concrete
- others

These companies are using the basalt as raw materials in it concrete.

Bazalt Reserves

The reserve of the Tal Parma is 310 million ton, and in other areas the reserves undefined as mentioned in Natural Resources Authority reports (2006)

1.1 competitor's analysis

There are some competitors companies such as:

AlManaseer group/ AlMotatawerah for crushers at AlHallabat region which have 4 crushers in Qaa' Khanna covering an area of about 5 donum, the table 6 below shows the selling prices:

Table5 the price of ton at AlHallabat crusher.

Kind	Price (JD)/ton	Price (JD)/m	Density (kg/m ³)
Swailleh	1.913	2.850	1490
Foleyah	3.430	4.750	1385
Humusseh	3.430	4.750	1385
Adasyah	3.442	4.750	1380
Semsemyah	3.442	4.750	1380
Ramel seel	3.276	4.750	1450

Source: Energy and minerals regulatory commission

- AlSharq company for crushers and contracting in Qaa' Khanna area is established on 5 donum, the company obtained the industrial quarry license in 10/11/2013, where the production of the company in 2013-2014 is 3,000 ton and 40,000 ton respectively.
- Almasar Almotahedah company which have two crushers at Qaa' Khanna area with 5 donum.
- Galeb AbuGreas company/Qaa' Khanna
- Ahmad Skaffe company/Aldhuleel
- Jordainian company for rockwool manufacturing/Qaa' Khanna

Price analysis and pricing policy

The local price of one ton of basalt is 8-10 JD and the annual production for these companies is 1,000,000 ton- Energy and minerals regulatory commission.

Market share

Based on the existing of production of basalt ores there is a good chance to compete with 10% market share, so the proposed production will be 100,000 ton/year in average about 333.333 ton/day, assuming 300 working day 8 hours work per day.

Projected revenues

Table 6 Projected revenues during years 2018-2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Rough Basalt price	8	8	8	9	9	9	10	10	10	10
Smooth Basalt price	10	10	11	11	11	12	12	12	13	13
Medium Basalt price	9	9	10	10	10	10	11	11	11	12
Rough Basalt rate	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%
Smooth Basalt rate	35%	35%	35%	35%	35%	35%	35%	35%	35%	35%
Medium Basalt rate	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%
Total revenues	895000	921850	949506	977991	1007330	1037550	1068677	1100737	1133759	1167772

4. Technical analysis

The required human resources

Table 7 the required human resources for the project

Job title	No.
Job Title	
General Manager	1
Geologist	1
Accountant	1
Guard	2
Maintenance technician	1
Mining technician	1
Driver	4
Cleaners	1
Total	12

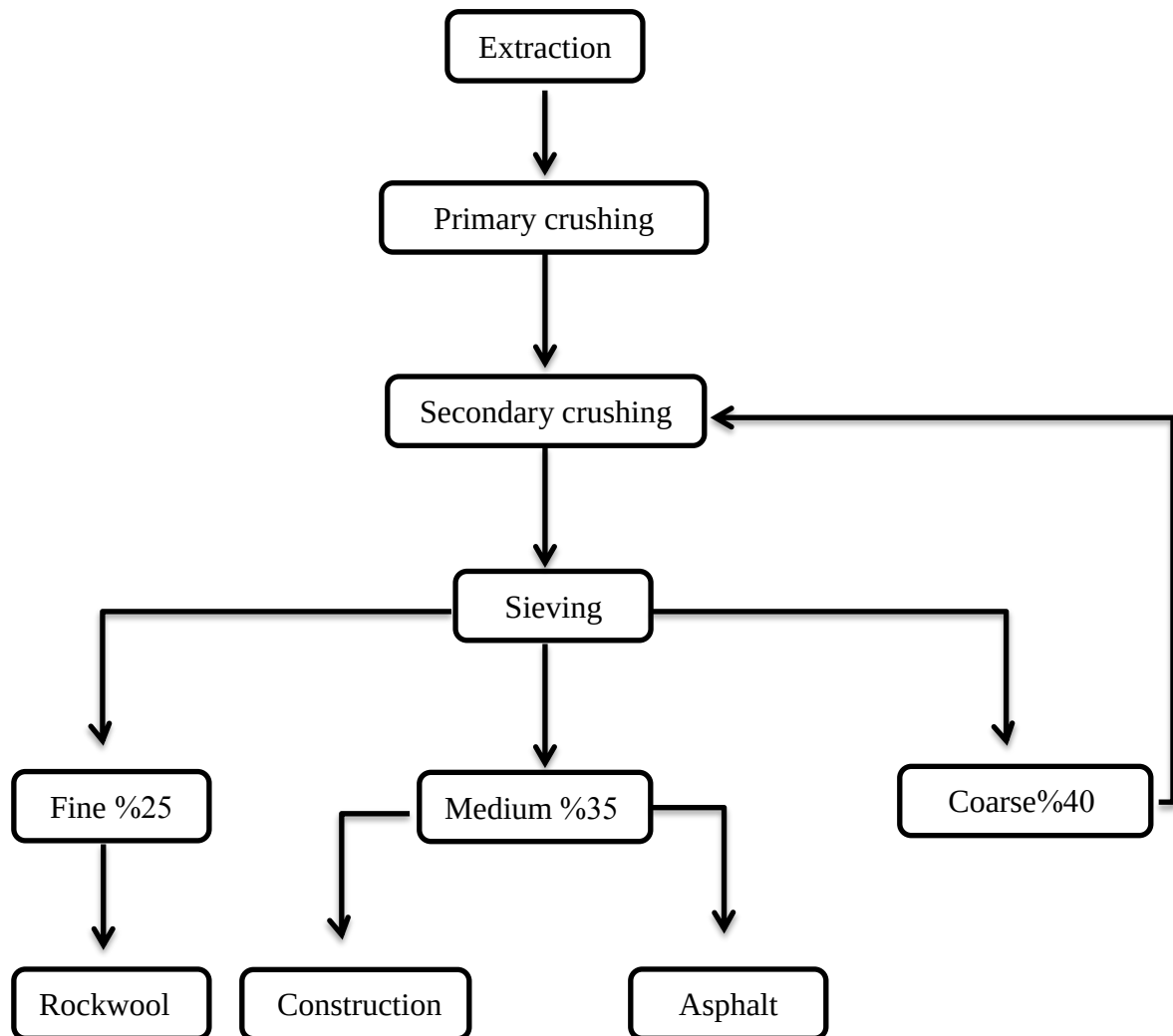
Site analysis

Based on Tal Parma geographic location which characterized being closing to high way road and ores deposited, where the reserves is about 310 million ton, the Fig.1 below shows the basalt ore deposits in Jordan

Figure 3 basalt ore



Production process and manufacturing steps



The exploiting of basaltic ores is starting with blasting then transported to crushing machines by vibrating feeder to remove the impurities from the basalt to be crushed using the jaw crusher and cone crusher to obtain the desired sizes .Finally, the sieving of ore will be carried out using vibrating screen to get an industrial product with different sizes (fine, medium, coarse) to be used in many industries.

Technical requirements analysis

1.1.1 Land and construction works

Assuming to rent a piece of land of with 35 donum, the following table shows the fees of mining requirements according to the recommendations of the Central Committee of State Property/ Energy and Minerals Regulatory Commission.

Table8 basalt mining fees

The fees	Mining fees (JD)
Fees of prospecting and exploration license of 1km ²	200
Fees of annual mining rights of 1km ²	500
Fees of prospecting license granting	500
Fees of rent for prospecting aspects about 1km ²	800
Fees of annual rent is 25 JD for one donum, 35 acres	875
Total	2,875

The following table shows the required construction works for the project:

Table 9 the required construction works

Construction work	Area (m ²)
Buildings	300
Paving	500

Machinery and Equipment

The extraction will be carried out by contracting companies specialized of extracting. The one ton of basalt ore cost 7 JD FOB to the factory about 2-5 km distance, the daily factory production will be 333.333 ton/day, and the total cost of daily ores transportation is (2,333) JD, and (699,600) JD per annum.

The following table shows the required machinery and equipment needed for the project:

Table 10 Machinery and equipments

Machinery and equipment	Quantity	Cost (JD per unit)
Loader	2	60,000
Crushing machine	2	40,000
Sieving machine	2	15,000

Machinery and equipment	Quantity	Cost (JD per unit)
Dumper	1	400,000
Pick up	2	23,000
Total	9	---

- Primary crushing machine :

Jaw crusher: Capacity (t/h) = 30-50, output size (mm) =10-40

Cone crusher: |Capacity (t/h):=30-50, Output size (mm) =<10mm

- Sieving machine :

Sieving machine: capacity $m^3/h = 30-50$, frequency (r/m) 970, motor power (kW) 45.

Furniture

The project needs office furniture (computers, tables and chairs, carpet, meeting requirements) that cost 12,000 JD

Vehicles and modes of transportation

The following table shows a list of vehicles and modes of transportation required for the project

Table 11 Vehicles and modes of transportation

Vehicles and modes of transportation	Quantity	Unit cost (JD)
Pickup truck	2	23,400

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees' Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	50.00%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table 16 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	20%
Income Tax Deduction(2)	0.50
Income Tax after Deduction	20%
Compulsory Reserve Percentage(3)	10%
Other Assumptions	
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premium (3)	10.12%
Income Tax Rate	20.00%
Beta (4)	0.89
Risk Premium	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 17 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	395,396
Loans Value	593,094
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	593,094
Loans Percentage	%60
Equity	
Cost of Equity	%15.5
Equity Value	395,396
Equity Percentage	%40
Gross	
Project Value	1,039,204
Proportion of the project	%100
Weighted Average Cost of Capital	10.75%

Capital Expenditures

The estimated cost of the project is 1,039,204 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 18 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Buildings	200	300	60,000
Paving	15	500	7,500
Other			6,750
Total			74,250

Table 19 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Loader	2	60,000	120,000
Crushing machine	1	40,000	40,000
Cone Crusher	1	40,000	40,000
Sieving machine	2	25,000	50,000
Total (JD)			250,000

Table 20 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	12,000
Total			12,000

Table 21 Transpiration and Vehicles Capex

transportation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,000	46,000
Truck	1	400,000	400,000
Total			446,000

Table 22 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	500

Transportation	750
Marketing	500
Training	750
Total	2500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 23 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	134,138
General and administrative expenses	3	%25	4,483
Marketing expenses	3	%25	559
Indirect Salaries	3	%25	20,800
Total			159,981

Table 24 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	74,250	10%	81,675	%7.9
Machinery & Equipment	250,000	10%	275,000	%26.5
Furniture	12,000	10%	13,200	%1.3
Vehicles	446,000	10%	490,600	%47.2
Pre-Operating Expenses	2,500	10%	2,750	%0.3
Working capital	159,981	10%	175,979	%16.9
Total (JD)	944,731		1,039,204	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,039,204) JD, where 60% of the project will be financed through loans (623,522) JD, and the remaining (40%) through (Equity) with (415,682) JD. The following table summarizes the general structure of the required financing:

Table 25 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	415,682	%40
Loans	623,522	%60
Total	1,039,204	%100
Use of Fund	Value	Ratio
Capital Expenditures	860,475	%82.80
Pre-Operating Expenses	2,750	%0.26
Working Capital	175,979	%16.93
Total	1,039,204	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 26 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	447,500	460,925	474,753	488,995	503,665	518,775	534,338	550,369	566,880	583,886
Utilities from	22,375	23,046	23,738	24,450	25,183	25,939	26,717	27,518	28,344	29,194
Maintenance from	8,950	9,219	9,495	9,780	10,073	10,376	10,687	11,007	11,338	11,678
Disposables from	8,950	9,219	9,495	9,780	10,073	10,376	10,687	11,007	11,338	11,678
Others	48,778	50,241	51,748	53,300	54,900	56,546	58,243	59,990	61,790	63,644
Total	536,553	552,649	569,229	586,305	603,895	622,011	640,672	659,892	679,689	700,079

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 27 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Transportaton	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rental expenses	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	6,454	6,647	6,847	7,052	7,264	7,481	7,706	7,937	8,175	8,420
Miscellaneous	1,630	1,679	1,730	1,782	1,835	1,890	1,947	2,005	2,065	2,127
Total	17,934	18,472	19,026	19,597	20,185	20,790	21,414	22,056	22,718	23,400

Marketing Expenses

Table 28 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	2,238	2,305	2,374	2,445	2,518	2,594	2,672	2,752	2,834	2,919

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 29 Expected monthly salaries (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	3,589
Geologist	550	573	596	620	646	672	700	729	759	2,154
Accountants	400	416	433	451	470	489	509	530	552	1,723
Mining technician	350	364	379	395	411	428	445	464	483	646
Maintenance technician	350	364	379	395	411	428	445	464	483	1,723
Cleaners	250	260	271	282	294	306	318	331	345	646
Driver	400	416	433	451	470	489	509	530	552	933
Guard	300	312	325	338	352	367	382	397	414	646
Workers	300	312	325	338	352	367	382	397	414	502

Table 30 Annual Expected Salaries

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Geologist	8,800	9,161	9,536	9,927	10,334	10,758	11,199	11,658	12,136	12,634
Accountants	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Mining technician	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Maintenance technician	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cleaners	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Driver	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Guard	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Grand Total	83,200	86,611	90,162	93,859	97,707	101,713	105,883	110,225	114,744	119,448

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 31 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	81,675	%5.0	%0.0
Machinaries	275,000	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	490,600	%10.0	%0.0
	860,475		

Table 32 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	860,475	866,635	872,938	879,388	885,989	892,744	899,659	906,737	913,984	921,403
Total Depreciation Expenses	81,964	82,580	83,210	83,855	84,515	85,191	85,882	86,590	87,315	88,057
Total Accumulated Depreciation	81,964	164,544	247,754	331,609	416,124	501,314	587,196	673,786	761,101	849,158

Total Additions	0	6,160	6,303	6,450	6,601	6,756	6,915	7,078	7,246	7,419
Total Net Book Values	778,511	702,092	625,184	547,779	469,865	391,430	312,462	232,951	152,883	72,245

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 33 Loan Details

Loan Value	623,522
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	162,304
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				623,522
2019	162,304	59,110	103,194	520,328
2020	162,304	49,327	112,977	407,351
2021	162,304	38,617	123,687	283,663
2022	162,304	26,891	135,413	148,250

2023

162,304

14,054

148,250

0

Income Statement

Table 34 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	895,000	921,850	949,506	977,991	1,007,330	1,037,550	1,068,677	1,100,737	1,133,759	1,167,772
Gross Operating Revenues	895,000	921,850	949,506	977,991	1,007,330	1,037,550	1,068,677	1,100,737	1,133,759	1,167,772
Operating Expenses	-536,553	-552,649	-569,229	-586,305	-603,895	-622,011	-640,672	-659,892	-679,689	-700,079
Gross Operating Profit	358,448	369,201	380,277	391,685	403,436	415,539	428,005	440,845	454,071	467,693
<i>Gross Profit Percentage</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>
Salaries and Benefits (Indirect Staff)	-83,200	-86,611	-90,162	-93,859	-97,707	-101,713	-105,883	-110,225	-114,744	-119,448
General and Administrative Expenses	-17,934	-18,472	-19,026	-19,597	-20,185	-20,790	-21,414	-22,056	-22,718	-23,400
Marketing Expenses	-2,238	-2,305	-2,374	-2,445	-2,518	-2,594	-2,672	-2,752	-2,834	-2,919
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-106,121	-107,388	-111,562	-115,901	-120,410	-125,097	-129,969	-135,033	-140,296	-145,767

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income before Interest, Depreciation and Tax	252,326	261,813	268,715	275,784	283,026	290,442	298,036	305,812	313,774	321,925
Fixed Assets Depreciations	-81,964	-82,580	-83,210	-83,855	-84,515	-85,191	-85,882	-86,590	-87,315	-88,057
Income before Tax and Interests	170,362	179,233	185,505	191,929	198,510	205,251	212,154	219,222	226,460	233,869
Bank Interests	-59,110	-49,327	-38,617	-26,891	-14,054	-0	-0	-0	-0	-0
Income Before Tax	111,252	129,906	146,888	165,038	184,456	205,251	212,154	219,222	226,460	233,869
Income Tax	-72,250	-75,981	-79,378	-83,008	-86,891	-91,050	-92,431	-93,844	-95,292	-96,774
Net Profit	39,002	53,925	67,510	82,031	97,565	114,201	119,723	125,378	131,168	137,095
Net Profit Percentage	%4	%6	%7	%8	%10	%11	%11	%11	%12	%12
Compulsory Reserves	-3,900	-5,393	-6,751	-8,203	-9,757	-11,420	-11,972	-12,538	-13,117	-13,709
Retained Earnings	35,102	83,634	144,394	218,221	306,030	408,810	516,561	629,401	747,452	870,838

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 35 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	39,002	53,925	67,510	82,031	97,565	114,201	119,723	125,378	131,168	137,095

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-monetary adjustments										
Bank benefits	59,110	49,327	38,617	26,891	14,054	0	0	0	0	0
Consumption	81,964	82,580	83,210	83,855	84,515	85,191	85,882	86,590	87,315	88,057
Operating cash flows before changes in working capital	180,076	185,832	189,337	192,777	196,134	199,391	205,605	211,968	218,482	225,152
Shortage (increase) in inventory	-37,292	-1,119	-1,152	-1,187	-1,222	-1,259	-1,297	-1,336	-1,376	-1,417
(Decrease) increase in accounts payable	14,842	445	459	472	487	501	516	532	548	564
Change in working capital	-22,450	-673	-694	-715	-736	-758	-781	-804	-828	-853
Net cash flows (used) from operating activities	157,626	185,158	188,644	192,062	195,398	198,633	204,824	211,164	217,654	224,298
Cash flows from investing activities										
(Purchase) of fixed assets	-860,475	-6,160	-6,303	-6,450	-6,601	-6,756	-6,915	-7,078	-7,246	-7,419
Net flows (used) from investment activities	-860,475	-6,160	-6,303	-6,450	-6,601	-6,756	-6,915	-7,078	-7,246	-7,419
Cash flows from financing activities										
Equity capital	415,682									
Loan Amortization	-103,194	-112,977	-123,687	-135,413	-148,250	0	0	0	0	0
Bank Interest Rate	-59,110	-49,327	-38,617	-26,891	-14,054	-0	-0	-0	-0	-0
Loans	623,522	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	876,900	-162,304	-162,304	-162,304	-162,304	0	0	0	0	0
Net increase (decrease) in cash	174,051	16,694	20,036	23,308	26,493	191,878	197,910	204,085	210,408	216,879

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows at the Beginning of Period	0	174,051	190,745	210,781	234,089	260,583	452,460	650,370	854,455	1,064,863
Cashflows at the End of Period	174,051	190,745	210,781	234,089	260,583	452,460	650,370	854,455	1,064,863	1,281,742

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 36 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	174,051	190,745	210,781	234,089	260,583	452,460	650,370	854,455	1,064,863	1,281,742
Inventory	37,292	38,410	39,563	40,750	41,972	43,231	44,528	45,864	47,240	48,657
Total assets traded	211,342	229,155	250,344	274,839	302,555	495,692	694,898	900,319	1,112,103	1,330,399
Non-current assets										
Fixed assets (net)	778,511	702,092	625,184	547,779	469,865	391,430	312,462	232,951	152,883	72,245
Total non-current assets	778,511	702,092	625,184	547,779	469,865	391,430	312,462	232,951	152,883	72,245
Total assets	989,854	931,247	875,528	822,618	772,419	887,121	1,007,361	1,133,270	1,264,985	1,402,644
Liabilities	-	-	-	-	-	-	-	-	-	-

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Current liabilities										
Payables	14,842	15,287	15,746	16,218	16,705	17,206	17,722	18,254	18,802	19,366
Remaining amount of Loan	112,977	123,687	135,413	148,250	0-	0-	0-	0-	0-	0
Total current liabilities	127,819	138,975	151,159	164,469	16,705	17,206	17,722	18,254	18,802	19,366
Non-current liabilities										
long term loans	407,351	283,663	148,250	0	0	0	0	0	0	0
Total long - term liabilities	407,351	283,663	148,250	0	0	0	0	0	0	0
Total liabilities	535,170	422,638	299,409	164,469	16,705	17,206	17,722	18,254	18,802	19,366
Property rights										
Member Contributions	415,682	415,682	415,682	415,682	415,682	415,682	415,682	415,682	415,682	415,682
Compulsory reserve	3,900	9,293	16,044	24,247	34,003	45,423	57,396	69,933	83,050	96,760
Profit (loss) round	35,102	83,634	144,394	218,221	306,030	408,810	516,561	629,401	747,452	870,838
Total equity	454,684	508,609	576,119	658,149	755,715	869,915	989,638	1,115,016	1,246,184	1,383,279
Total liabilities and equity	989,854	931,247	875,528	822,618	772,419	887,121	1,007,361	1,133,270	1,264,985	1,402,644

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 37 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-1,039,204	160,376	178,998	182,341	185,613	188,798	191,878	197,910	204,085	210,408	216,879	3,340,001
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash	-1,039,204	118,052	118,967	109,422	100,571	92,364	84,757	78,934	73,494	68,414	63,672	980,562

Table below (38) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 38 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,039,204	160,376	178,998	182,341	185,613	188,798	191,878	197,910	204,085	210,408	216,879
Project Value Returned	1,039,204	878,828	699,830	517,489	331,877	143,079	-48,799	-246,709	-450,794	-661,201	-878,081
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 39 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	850,004

Payback Period	5
Internal Rate of Return	%22.98

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 850,004 Jordan Dinars considering that the project provides 11 Job Opportunities for the governorate residents.

Table 40 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%23.87	5	10.75%
Revenues declined by 10%	%20.65	6	10.75%
Operating Expenses Increased by 10%	%18.41	7	10.75%